

Smart Parking Management and Slot Monitoring System using Verilog

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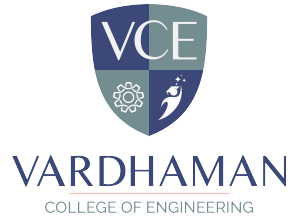
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CERTIFICATE

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Abstract

This project presents the design, simulation, and functional verification of a Smart Parking Management and Slot Monitoring System implemented using Verilog Hardware Description Language (HDL). With the rapid growth of vehicle ownership in urban areas, manual parking management has become inefficient, time-consuming, and prone to errors, motivating the need for a digital slot-monitoring solution. The proposed system models an eight-slot parking facility using an 8-bit occupancy register, in which each bit represents the real-time status (occupied or vacant) of an individual parking slot.

The design is implemented as a synchronous sequential circuit that is driven by a system clock and an active-high asynchronous reset. Two control inputs, *entry* and *exit*, together with a 3-bit slot-select input (*slot_no*), allow a vehicle to be allocated to, or released from, any one of the eight available slots. On assertion of the *entry* signal, the corresponding bit of the *slots* output register is set to indicate occupancy, while assertion of the *exit* signal clears the corresponding bit to indicate vacancy. The reset signal initializes the entire slot register to a known, fully-vacant state, ensuring predictable start-up behaviour.

A dedicated self checking testbench was developed to verify the functionality of the design under various operating scenarios, including reset behaviour, sequential entry of vehicles into different slots, and vehicle exit. The design was simulated, and the resulting waveforms were analysed to confirm correct read/write behaviour of the slot register, proper synchronization with the clock, and accurate reflection of slot occupancy for every entry and exit event. The simulation results validate that the proposed architecture correctly implements real-time parking slot allocation and monitoring using a compact and easily extensible RTL design, making it a suitable foundation for scaling to larger parking facilities or for future integration with IoT-based monitoring and display systems.

Keywords: Smart Parking System, Verilog HDL, Slot Monitoring, RTL Design, Digital Logic Design, Sequential Circuit, Functional Verification, Testbench Simulation.

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Abbreviations

Abbreviation	Description
HDL	Hardware Description Language
RTL	Register Transfer Level
IoT	Internet of Things
FPGA	Field Programmable Gate Array
FSM	Finite State Machine
IR	Infrared
LED	Light Emitting Diode
VCD	Value Change Dump
LUT	Look-Up Table
I/O	Input / Output

CHAPTER 1

Introduction

1.1 Introduction

The exponential growth of vehicle ownership in urban and semi-urban areas has placed increasing pressure on the availability and management of parking infrastructure. In conventional parking facilities, slot allocation is largely managed manually by security personnel or through simple barrier-based systems that provide no real-time information regarding slot occupancy. This often leads to wasted time, fuel wastage, unnecessary vehicular congestion near entry points, and an overall poor user experience. To address these issues, digital and automated parking management solutions have gained significant attention in both industry and academia.

Hardware-based digital design using Hardware Description Languages (HDL) such as Verilog offers an efficient and reliable approach to implement real-time monitoring and control logic for parking systems. Unlike software-only or microcontroller-based approaches, an HDL-based design can be directly synthesized onto Field Programmable Gate Arrays (FPGAs) or Application-Specific Integrated Circuits (ASICs), providing deterministic timing, low latency, and the potential for tight integration with sensor interfaces such as infrared (IR) or ultrasonic proximity sensors. This makes Verilog HDL a suitable choice for building the core slot-monitoring and allocation logic of a smart parking system.

1.2 Smart Parking Systems

A smart parking system is broadly defined as any parking management solution that automatically senses, records, and communicates the occupancy status of individual parking slots, thereby enabling efficient slot allocation and utilization. Such systems typically consist of three functional layers: a

sensing layer (vehicle detection at each slot using sensors), a control/logic layer (which processes entry and exit events and updates slot status), and a display/communication layer (which presents the current occupancy status to users or a central monitoring system).

The proposed work in this project focuses primarily on the control/logic layer, which forms the digital core of the parking system. This layer is responsible for accurately tracking the occupancy of each parking slot based on entry and exit events, ensuring that the system state remains consistent and correctly reflects real-world conditions at all times.

1.3 Verilog HDL for Digital System Design

Hardware Description Languages such as Verilog enable designers to describe digital hardware behaviour at the register-transfer level (RTL), which can subsequently be simulated for functional correctness and synthesized into physical hardware. Verilog HDL supports both behavioural and structural modelling styles, making it well suited for describing sequential logic such as registers, counters, and finite state machines that are commonly used in slot-monitoring and control applications.

Using Verilog HDL, a parking slot occupancy register can be modelled as a simple, synchronous, edge-triggered sequential circuit that updates its state on every active clock edge based on control inputs. This RTL description can be verified extensively through simulation using a self-checking testbench prior to physical implementation, ensuring functional correctness and reducing the risk of hardware-level errors after fabrication or FPGA deployment.

1.4 Objectives of the Project Work

- Design a digital Smart Parking Management and Slot Monitoring System using Verilog HDL that can track the occupancy status of multiple parking slots in real time.
- Model an 8-slot parking facility using an 8-bit occupancy register, with

each bit representing the vacant/occupied status of a single slot.

- Design the control logic required to update slot occupancy on assertion of *entry* and *exit* signals, using a 3-bit slot-select input to address the required slot.
- Implement a synchronous, clock-driven architecture with an asynchronous active-high reset that initializes all slots to the vacant state.
- Develop a comprehensive Verilog testbench to functionally verify the design under various operating conditions, including reset, sequential vehicle entries into different slots, and vehicle exit.
- Perform functional verification of the design using simulation and waveform analysis to confirm correct slot allocation and de-allocation behaviour.
- Ensure that the design is modular and easily scalable to a larger number of parking slots, and that it can serve as a foundation for future integration with sensor interfaces and display/communication modules.

CHAPTER 2

Literature Survey

2.1 Introduction

The literature on automated and smart parking systems has grown considerably over the past two decades, driven by increasing urban vehicle density and the need for efficient space utilization. Researchers have approached the problem from multiple perspectives, including FPGA/HDL-based hardware controllers, microcontroller and IoT-based sensing systems, and vision/machine-learning-based occupancy detection. This chapter reviews recent and relevant studies to understand existing design practices and to identify the research gap addressed by the proposed work.

2.2 Review of Prior Research

A significant body of work focuses on FPGA and Verilog/VHDL-based implementations of parking slot management logic. Kaur and Singh presented an early design and implementation of a car parking system on FPGA, demonstrating that finite-state-machine-based control logic can reliably manage slot allocation using hardware description languages. Song et al proposed an intelligent parking system design using FPGA, highlighting the advantages of hardware-based control in terms of deterministic timing and low-latency response to sensor events. More recent works have extended these ideas: secure parking systems combining Verilog-based slot allocation with password-protected entry and exit, implemented on modern Artix-7 FPGA boards, have been reported to provide both security and accurate duration-based billing. Similarly, intelligent multi-slot parking systems using finite state machines synthesized with industry-standard tools such as Xilinx ISE have demonstrated low standby power consumption and efficient use of FPGA resources.

Beyond FPGA only approaches, several studies extend parking management

systems with additional sensing and I/O features. A multi-car parking system using Verilog HDL was proposed to accommodate several vehicles simultaneously while maintaining a modular, easily extensible RTL structure. Extended designs incorporating Moore-based finite state machines together with sensors and electric security locks have also been proposed for secured car parking, using languages such as VHDL alongside supporting hardware components.

In parallel with hardware-centric approaches, IoT-based smart parking systems have received substantial attention. These systems typically employ ultrasonic or infrared sensors interfaced with microcontrollers such as the ESP32 or NodeMCU to detect slot occupancy and relay this information to a cloud server or mobile application for real-time monitoring. Comprehensive real-time parking control and monitoring systems have also incorporated camera-based sensing and software analytics to optimize distribution of parking slots and provide additional functionalities such as car localization and automated reporting.

2.2.1 Identified Research Gaps

While extensive research exists on both FPGA/HDL-based parking controllers and IoT/sensor-based smart parking systems, most FPGA-based studies emphasize complex, feature-rich designs (e.g., password protection, billing, or multi-sensor fusion) without adequately documenting a simple, well-verified baseline RTL model for slot occupancy tracking that can serve as an easily extensible foundation. Additionally, many IoT-based studies focus on the sensing and communication layers while treating the underlying digital control logic superficially. Furthermore, verification details such as the specific test scenarios used to validate correctness are rarely documented in detail, and scalability to a larger number of slots is seldom discussed as an explicit design goal. This indicates the need for a clean, well-verified Verilog HDL implementation of the core slot-monitoring logic one that can be independently validated through simulation and subsequently extended with sensor interfaces, security features, or IoT connectivity as required.

2.3 Summary

This literature survey reviewed recent advancements in FPGA/HDL-based parking controllers and IoT-based smart parking systems. Prior research demonstrates that finite-state-machine and register-based control logic implemented in Verilog or VHDL can reliably manage parking slot allocation, while IoT-based systems extend this functionality with real-time remote monitoring. The identified research gap underscores the necessity for a clearly verified, modular Verilog HDL model of parking slot occupancy tracking, which forms the motivation for the proposed work presented in this report.

CHAPTER 3

Existing Approaches to Smart Parking Systems

3.1 Introduction

Existing smart parking solutions can broadly be classified into three categories based on their underlying implementation technology: (i) FPGA/HDL-based hardware controllers, (ii) microcontroller and IoT-based sensing systems, and (iii) vision or machine-learning-based occupancy detection systems. Each of these approaches offers different trade-offs in terms of cost, response time, scalability, and complexity of implementation.

3.2 Detailed Description of Existing Techniques

3.2.1 FPGA and HDL-Based Parking Controllers

FPGA-based parking systems implement the slot allocation and control logic directly in hardware using Verilog or VHDL. Kaur and Singh demonstrated an early car parking system design on FPGA using structured HDL modules for slot control. Song et al. implemented an intelligent parking controller using fuzzy-logic-assisted hardware, showing the benefit of dedicated hardware for reduced computation latency compared to purely software-based controllers. More recent secure parking implementations combine slot allocation logic with password-protected access control and duration-based billing, synthesized on Artix-7 class FPGA devices. These works confirm that Moore or Mealy finite-state-machine models, together with simple occupancy registers, are well suited for representing parking slot states in hardware.

3.2.2 Microcontroller and IoT-Based Sensing Systems

A large class of existing systems relies on microcontrollers such as the Arduino, ESP32, or NodeMCU interfaced with ultrasonic or infrared sensors

placed at individual parking slots. These systems detect vehicle presence at each slot and transmit occupancy data over Wi-Fi to a cloud backend or mobile application, allowing users to check slot availability remotely. Such designs typically emphasize connectivity and user-facing features (e.g., QR-code-based access, mobile dashboards) rather than the internal digital logic used to track occupancy, which is often implemented in simple firmware rather than dedicated hardware description logic.

3.2.3 Vision and Analytics Based Systems

More advanced parking management systems integrate camera-based occupancy detection combined with software analytics and machine learning models to optimize slot allocation dynamically, track vehicle dwell time, and generate automated reports. While powerful, such systems require significantly more computational resources and are generally implemented as software layers running on general-purpose processors rather than as dedicated digital hardware.

3.3 Motivation for the Proposed Work

Across all three categories of existing work, the fundamental requirement remains the same: a reliable mechanism to track the real-time occupancy of each individual parking slot and to correctly update this status whenever a vehicle enters or exits. Many existing FPGA-based designs bundle this core logic with additional features such as security, billing, or multi-sensor fusion, making it harder to isolate, understand, and verify the essential slot-monitoring behaviour in isolation.

The proposed work therefore focuses on designing and thoroughly verifying a compact, modular Verilog HDL model that implements this essential occupancy-tracking function using an 8-bit slot-status register, controlled through simple *entry*, *exit*, and *slot_no* signals. This baseline design is intended to be easily understandable, simulate correctly under all expected operating conditions, and serve as a solid foundation onto which sensor interfaces, security features, or IoT connectivity can later be added.

3.4 Summary

Existing work on smart parking systems demonstrates a strong foundation in FPGA/HDL-based control logic, IoT-based sensing and communication, and vision-based analytics. However, many implementations bundle the core slot-monitoring logic with additional application-specific features, making the underlying control logic harder to isolate and verify independently. This motivates the proposed design and verification of a clean, modular Verilog HDL model for parking slot occupancy tracking, presented in the next chapter.

CHAPTER 4

RTL Design and Verification of the Smart Parking Management System in Verilog HDL

4.1 Introduction

The proposed work presents the design, implementation, and functional verification of a Smart Parking Management and Slot Monitoring System using Verilog HDL. The system models an 8-slot parking facility in which the occupancy status of each slot is represented by a single bit of an 8-bit output register, *slots[7:0]*. The primary objective is to design a compact, synchronous digital circuit that accurately tracks slot occupancy in real time based on vehicle entry and exit events.

The proposed architecture is fully synchronous, updating its internal state on the positive edge of the system clock, and includes an asynchronous, active-high reset that initializes all parking slots to the vacant (logic 0) state. Two control signals, *entry* and *exit*, together with a 3-bit slot address input, *slot_no[2:0]*, allow any one of the eight slots to be set to the occupied state or cleared to the vacant state on a given clock edge. By implementing the system at the RTL level, the design provides a clear, synthesizable, and easily verifiable model of the core logic required for real-time parking slot management.

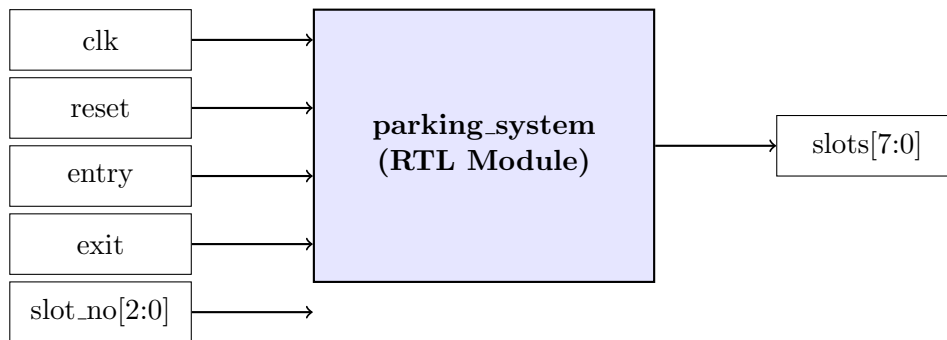
4.2 Proposed System Architecture

The Smart Parking Management System is implemented as a single Verilog module, `parking_system`, which integrates the slot-occupancy register together with the control logic required to update it. Table 4.1 summarizes the input and output ports of the design.

Figure 4.1 illustrates the top-level block diagram of the proposed design.

Table 4.1: Input/Output Ports of the Parking System Module

Port	Direction	Width	Description
clk	Input	1 bit	System clock; state updates occur on the positive edge.
reset	Input	1 bit	Active-high asynchronous reset; clears all slots to vacant.
entry	Input	1 bit	Asserted when a vehicle enters the selected slot.
exit	Input	1 bit	Asserted when a vehicle exits the selected slot.
slot_no	Input	3 bits	Binary address (0–7) of the slot being accessed.
slots	Output	8 bits	Register holding the occupancy status of all 8 slots.

**Figure 4.1:** Block Diagram of the Smart Parking Management System

The *clk* and *reset* signals control the synchronous behaviour of the internal 8-bit register, while *entry*, *exit*, and *slot_no* together determine which bit of the *slots* register is modified on a given clock edge. The *slots[7:0]* output continuously reflects the real-time occupancy status of all eight parking slots and can be directly interfaced with display logic, LED indicators, or a higher-level monitoring/communication module in a complete parking management system.

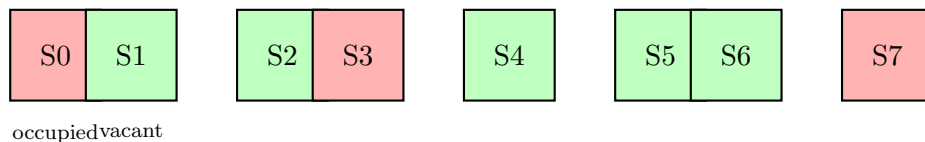
**Figure 4.2:** Example Snapshot of the 8-Slot Occupancy Register (*slots[7:0]*)

Figure 4.2 depicts an example snapshot of the parking facility, where each of the eight slots (S0–S7) corresponds directly to one bit of the *slots[7:0]* register. A slot shown in red (logic 1) indicates that the slot is currently

occupied, while a slot shown in green (logic 0) indicates that it is vacant. This one-to-one mapping between register bits and physical slots makes the design intuitive to interface with external indicator hardware such as LEDs.

4.3 Verilog HDL Implementation

The core functionality of the design is implemented using a single always block that is sensitive to the positive edge of *clk* and the positive edge of *reset*, implementing an asynchronous reset with synchronous update behaviour, as shown below.

```
module parking_system(  
    input clk,  
    input reset,  
    input entry,  
    input exit,  
    input [2:0] slot_no,  
    output reg [7:0] slots  
);  
  
always @(posedge clk or posedge reset)  
begin  
    if(reset)  
        slots <= 8'b00000000;  
  
    else begin  
        if(entry)  
            slots[slot_no] <= 1'b1;  
        if(exit)  
            slots[slot_no] <= 1'b0;  
    end  
end  
endmodule
```

When *reset* is asserted, the entire *slots* register is immediately cleared to 8'b00000000, representing a fully vacant parking facility, regardless of the clock edge. Once *reset* is de-asserted, the register updates only on the rising edge of *clk*. If *entry* is asserted, the bit of *slots* addressed by *slot_no* is set to 1, marking that slot as occupied. If *exit* is asserted, the addressed bit is cleared to 0, marking the slot as vacant. Because *slot_no* is used as a bit-select index into the 8-bit register, the design naturally supports random access to any of the eight slots without requiring a separate decoder or multiplexer structure, resulting in a compact and efficient RTL description.

4.4 Verification Methodology and Setup

To ensure the correctness and reliability of the proposed Smart Parking Management System, a dedicated Verilog testbench was developed and simulated.

4.4.1 Simulation-Based Verification

The primary verification method used was functional simulation of the `parking_system` module using a self-checking Verilog testbench. The testbench, named `test`, instantiates the design under test as `uut` and drives it with a periodic clock generated using an `always #5 clk = ~clk;` statement, corresponding to a 10 ns clock period (100 MHz). All output and internal signals were dumped to a Value Change Dump (VCD) file for waveform analysis.

4.4.2 Verification Approach

- **Reset Behaviour:** Verified that asserting *reset* clears the *slots* register to 8'b00000000 regardless of prior state.
- **Slot Entry:** Verified that asserting *entry* with a given *slot_no* correctly sets only the corresponding bit of *slots*, leaving all other bits unaffected.

- **Slot Exit:** Verified that asserting *exit* with a given *slot_no* correctly clears only the corresponding bit of *slots*.
- **Multiple Slot Operations:** Verified correct, independent operation across different slot addresses (e.g., slot 0 and slot 3) without interference between slots.
- **Timing/Synchronization:** Verified that all state updates occur synchronously with the rising edge of the clock, and that the design responds correctly to the asynchronous reset.

The testbench sequence first applies a reset pulse, then issues an entry event for slot 0, followed by an entry event for slot 3, and finally an exit event for slot 0, allowing the correctness of both the setting and clearing of individual bits within the 8-bit occupancy register to be verified across the full duration of the simulation.

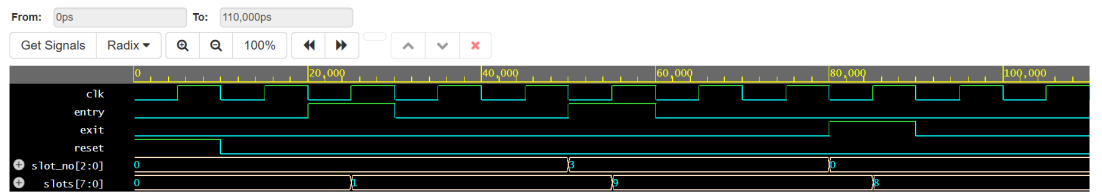


Figure 4.3: Simulation Waveform of the Smart Parking Management System

Figure 4.3 shows the simulation waveform obtained during functional verification of the design. The waveform captures the *clk*, *entry*, *exit*, *reset*, *slot_no[2:0]*, and *slots[7:0]* signals over the simulation duration. Initially, *reset* is held high, and the *slots* register remains at 0. Once *reset* is de-asserted, an *entry* event with *slot_no* = 0 causes *slots* to change from 0 to 1 (binary 00000001), indicating that slot 0 has become occupied. Subsequently, an *entry* event with *slot_no* = 3 updates *slots* to 9 (binary 00001001), showing that slot 3 has also become occupied while slot 0 remains occupied. Finally, an *exit* event with *slot_no* = 0 updates *slots* to 8 (binary 00001000), confirming that slot 0 has correctly returned to the vacant state while slot 3 remains occupied.

This behaviour confirms that the design correctly and independently tracks

the occupancy of each parking slot in response to *entry* and *exit* events, and that the asynchronous reset reliably initializes the system to a known state.

4.5 Summary

This chapter presented the proposed RTL architecture of the Smart Parking Management and Slot Monitoring System, along with its Verilog HDL implementation and verification methodology. The design uses a compact 8-bit occupancy register controlled through *entry*, *exit*, and *slot_no* signals, with an asynchronous active-high reset. Functional verification was carried out using a dedicated Verilog testbench, and the resulting simulation waveforms confirmed correct slot allocation, de-allocation, and reset behaviour. The next chapter presents a detailed discussion of the simulation results.

CHAPTER 5

Results and Discussion

5.1 Introduction

This chapter presents the functional verification outcomes of the proposed Smart Parking Management and Slot Monitoring System. The design was simulated using the developed Verilog testbench, and the resulting waveforms were analysed to validate correct behaviour of the reset logic, slot entry logic, and slot exit logic. This section examines the observed signal behaviour in detail and correlates it with the expected functionality of the design as per the RTL specification presented in Chapter 4.

5.2 Overview of Results

The simulation of the `parking_system` module demonstrated correct functional operation across all applied test scenarios. The design responded correctly to the asynchronous reset, updated the appropriate bit of the *slots* register on every *entry* and *exit* event, and maintained the state of all other slots unaffected during each operation. No unintended bit toggling or race conditions were observed in the simulation output.

Table 5.1 summarizes the test scenarios applied through the testbench, the stimulus applied at each stage, and the corresponding value observed on the *slots*[7:0] output.

Table 5.1 confirms that the observed values of the *slots* register at each stage of the simulation match the expected values derived from the RTL specification, validating the correctness of the design under the applied test sequence.

Table 5.1: Summary of Test Cases and Simulation Results

Test #	Stimulus Applied	slot_no	Observed slots[7:0]
1	reset = 1 (initial state)	–	8'b00000000
2	reset de-asserted; entry = 1	0	8'b00000001 (slot 0 occupied)
3	entry de-asserted; entry = 1	3	8'b00001001 (slots 0 and 3 occupied)
4	entry de-asserted; exit = 1	0	8'b00001000 (slot 0 vacated, slot 3 remains occupied)

5.3 Analysis and Interpretation

The simulation waveform in Figure 4.3 (Chapter 4) provides clear evidence of correct operation of the Smart Parking Management System.

During the initial phase of the simulation, the *reset* signal is held high, and the *slots* register remains at its default value of 8'b00000000, confirming correct asynchronous reset behaviour. Once *reset* is de-asserted, the design correctly responds to the periodic clock and to changes in the *entry*, *exit*, and *slot_no* signals.

When *entry* is asserted with *slot_no* = 0, only bit 0 of the *slots* register transitions from 0 to 1; all other bits remain unaffected, confirming that the bit-select addressing implemented through `slots[slot_no]` operates correctly. When *entry* is subsequently asserted with *slot_no* = 3, bit 3 of the register transitions to 1 while bit 0 (already set) remains unchanged, demonstrating that the design correctly supports the independent occupancy of multiple slots simultaneously. Finally, when *exit* is asserted with *slot_no* = 0, only bit 0 is cleared back to 0, while bit 3 remains set, confirming correct and isolated clearing of individual slot status bits.

Table 5.2: Bit-to-Slot Mapping of the slots[7:0] Register

Bit Index	7	6	5	4	3	2	1	0
Slot ID	S7	S6	S5	S4	S3	S2	S1	S0

Table 5.2 illustrates the direct correspondence between each bit of the *slots*[7:0] output register and a physical parking slot, further reinforcing the simplicity and interpretability of the proposed design. This one-to-one mapping

enables the *slots* register to be directly connected to display hardware (e.g., eight LED indicators) without requiring any additional decoding logic.

The successful verification of reset behaviour, independent slot entry, and independent slot exit demonstrates that the proposed design meets its core functional requirements. Because state updates are synchronized to the system clock and the addressed slot bit is modified independently of all other bits, the design is inherently free from interference between simultaneously occupied slots, which is essential for reliable real-time parking slot management.

5.4 Summary

The Smart Parking Management and Slot Monitoring System was successfully designed in Verilog HDL and verified through simulation. The design correctly implements reset initialization, independent slot entry, and independent slot exit operations for all eight parking slots represented by the *slots*[7:0] register. Waveform analysis confirmed accurate, isolated bit-level updates for each entry and exit event, with no interference observed between different slots.

Overall, the results validate the correctness of the proposed RTL design and confirm that it satisfies the functional objectives defined for the project. The compact and modular nature of the design, together with the direct bit-to-slot mapping of the output register, makes it a strong foundation for future extension with sensor interfaces, display modules, and IoT-based remote monitoring capabilities.

CHAPTER 6

Conclusions and Future Scope

6.1 Conclusion

The design and verification of a Smart Parking Management and Slot Monitoring System using Verilog HDL effectively demonstrates how real-time parking slot occupancy can be tracked and managed using a compact, synchronous digital circuit. The proposed design models an 8-slot parking facility through an 8-bit occupancy register, in which each bit independently represents the vacant or occupied status of a corresponding parking slot. The control logic, driven by *entry*, *exit*, and *slot_no* signals, allows any individual slot to be marked as occupied or vacant on a given clock edge, while an asynchronous active-high reset ensures the system always initializes to a known, fully-vacant state.

Functional verification of the design was carried out using a dedicated Verilog testbench and simulation-based waveform analysis, confirming correct reset behaviour, accurate and isolated updates of individual slot-status bits during entry and exit events, and reliable synchronous operation under a periodic clock. The simulation results validate that the design meets all of its stated functional objectives.

This project provides valuable hands-on experience in digital system design, RTL modelling, and functional verification using Verilog HDL, while also reinforcing key concepts of sequential circuit design such as synchronous registers, asynchronous reset, and bit-level addressing. The resulting design serves as a clean, well-verified foundation for a complete smart parking solution and can be readily extended to incorporate additional real-world features.

6.2 Future Scope of Work

The designed Smart Parking Management and Slot Monitoring System serves as an excellent starting point for building a more complete, feature-rich parking solution. While the current design fulfills its core functional objective of accurate slot occupancy tracking, several enhancements can be made in future work.

One direction for future development is the integration of real sensor interfaces, such as infrared (IR) or ultrasonic proximity sensors, at each physical parking slot, replacing the simulated *entry* and *exit* stimulus signals with actual vehicle-detection hardware. This would allow the design to be deployed and tested on a real FPGA platform interfaced with physical parking infrastructure.

Further extensions could include the addition of a display module, such as a seven-segment display or LED matrix driven directly by the `slots[7:0]` register, to provide a visual representation of slot availability at the entrance of the parking facility. The design could also be extended to compute and display the total number of available slots using a simple population-count (bit-counting) circuit.

The system could also be extended with communication interfaces, such as UART or SPI, to transmit real-time slot-occupancy data to an external microcontroller or IoT gateway, enabling remote monitoring through a mobile application or cloud dashboard.

Additional future enhancements could include slot-level access control (e.g., password- or RFID-based entry authorization), automated billing based on parking duration using an internal timer/counter module, and scaling the design to support a larger number of slots or multiple parking levels using a parameterized, hierarchical architecture.

Lastly, the design can be synthesized and implemented on an actual FPGA platform to validate its real-time operation under practical timing and resource constraints. More advanced verification methodologies, such as SystemVerilog assertions or constrained-random testing.

Mapping of Program Outcomes (POs) and Sustainable Development Goals (SDGs)

1. Contribution to Program Outcomes (POs) and Program Specific Outcomes (PSOs)

The project contributes to the attainment of the following Program Outcomes:

PO1: Engineering Knowledge

This project demonstrates the application of fundamental knowledge in digital electronics, sequential circuit design, and Verilog HDL modelling. The design, analysis, and implementation phases required the use of core concepts related to synchronous logic, register-based state storage, and RTL-level verification, thereby fulfilling PO1.

PO2: Problem Analysis

The project involves identifying the functional requirements of a real-time parking slot monitoring system, including correct handling of concurrent entry and exit events and reset behaviour. Systematic analysis and debugging using simulation and testbench development address these challenges, thereby fulfilling PO2.

PO3: Design/Development of Solutions

The project focuses on designing a functional slot-monitoring digital circuit that meets specified correctness requirements for real-time occupancy tracking. The structured RTL design and verification flow using Verilog HDL demonstrates the ability to design solutions for practical engineering problems, thereby fulfilling PO3.

PO5: Engineering Tool Usage

Modern engineering tools such as HDL simulators and waveform analyzers are utilized for design and verification. The effective use of these tools for modelling, testing, and debugging fulfills PO5.

2. Program Specific Outcomes (PSOs)

PSO1: Apply the knowledge of domain-specific skill set for the design and analysis of components in VLSI and Embedded systems.

The project applies domain-specific knowledge in digital system design, RTL modelling using Verilog HDL, and functional verification techniques for the design and analysis of a Smart Parking Management and Slot Monitoring System, thereby satisfying PSO1.

3. Alignment with Sustainable Development Goals (SDGs)

The project supports the following Sustainable Development Goals:

SDG 9: Industry, Innovation and Infrastructure

The project directly aligns with SDG 9 by contributing to the development of intelligent digital infrastructure for parking management. The design and verification of a modular, hardware-based slot-monitoring system supports innovation in smart-city and smart-infrastructure applications.

SDG 11: Sustainable Cities and Communities

By enabling accurate, real-time tracking of parking slot availability, the proposed system helps reduce the time vehicles spend searching for parking, thereby reducing unnecessary fuel consumption, traffic congestion, and emissions in urban areas, in support of more sustainable and liveable cities.

SDG 12: Responsible Consumption and Production

The compact, resource-efficient RTL design minimizes hardware overhead while providing the essential functionality required for parking management, encouraging responsible and efficient use of digital hardware resources.

Overall Impact

The project “Smart Parking Management and Slot Monitoring System using Verilog” has significant academic, technical, and societal impact by addressing the growing need for efficient, automated parking infrastructure through open, verifiable digital hardware design.

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